Scope: University Course Registration System

Level: User Goal

Primary Actor: Department manager

Stakeholders and Interests:

- **Department:** Department is collection of students, lecturers, and teaching assistants. Each department can have their custom configuration file. Configuration file contains information about department, lab section capacity limit, elective course limit, given courses in that department and lecturers of department.
- **Curriculum:** Curriculum identifies which course can be taken in which semester. Students can not take courses from other semesters instead of current semester.
- **Courses:** Courses may in type of mandatory or elective. Also, elective courses have several types (non-technical, technical, faculty technical, university elective). A course can have one or more lab sections. Each lab section instructed by a teaching assistant and one course may have many lab sections. A course can have pre-requisite courses, or it can be prerequisite for another courses. Students can not take courses if they failed any pre-requisite of a course. Elective courses and lab sections has capacity limit which determined by department.
- **Lecturer:** Lecturer can be instructor of one or many courses. Also, lecturers advising to students. Lecturer may not have advised students.
- **Teaching Assistant:** Teaching assistant instructor of lab sections of department courses. Each lab section can have only one teaching assistant, but one course can have many lab sections.
- **Student:** Students needs to take courses in curriculum order. They can not take course from previous semesters or coming semesters. They can select courses which has collision, but these selections not approved by their advisors. If student needs to select an elective course or lab section these are selected from a poll by randomly.
- **Transcript:** Transcript shows us student's taken courses and their grades, completed credits, semester GPA and cumulative GPA.

Preconditions: Department manager is identified and authenticated.

Main Success Scenario:

1. Department manager starts the software
2. The software creates randomly generated students with given limit.
3. For each course taken by a student it assigns random point between 0-100. By using that grade it calculates letter grade and grade in [0, 4] format. For elective courses a random course will be selected from elective course pool. Also, for the lab sections a random section will be selected from lab sections.
4. After course registration process system will generate an output file for each student. That output file contains student information and 2 transcripts in JSON format. First transcript contains courses from previous semester, the second one contains all courses included newly taken courses.
5. After student output files generated the system will generate a output file for the department. It contains information about which course could not be taken by how many students.

Technology and Data Variations List:

- Each run of the software it will generate new random students.

Alternative Flows:

3a

If capacity of all elective courses is full student can not register any elective course.

3b

If lab section capacity is full for each section student can not take a lab section.

3c

If student has any collision (taken two or more courses with a same schedule, all lecture hours does not need to be the same) in his timetable advisor won't approve registration of the second course.

3d

If course grade less than 1.5 student fails course. And he/she can not take other courses if failed course is a prerequisite for them.